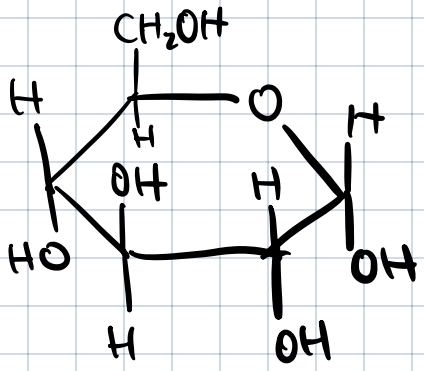
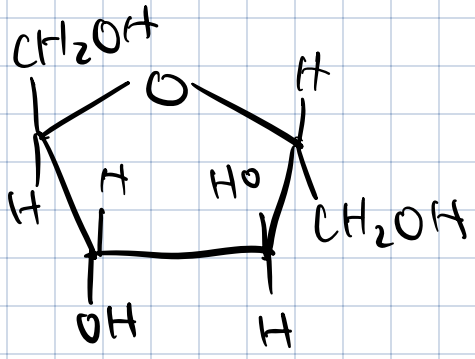
Glucose



aldol hexose

$C_6H_{12}O_6$

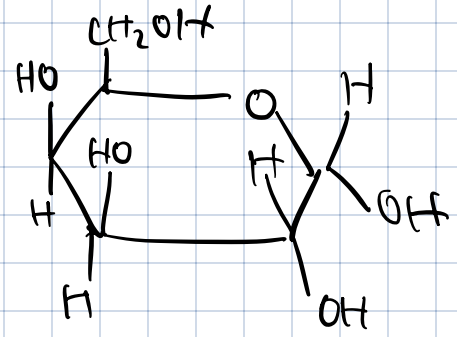
Fructose



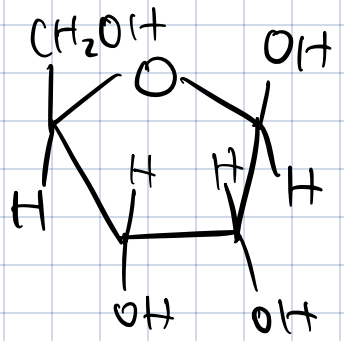
ketohexose

$C_6H_{12}O_6$

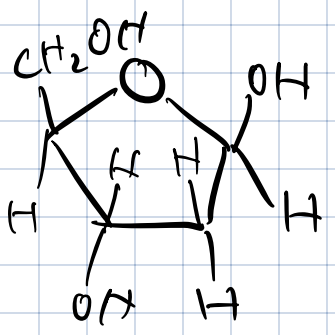
Galactose



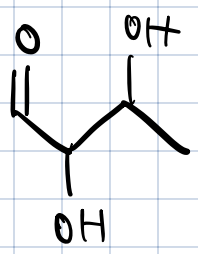
$C_6H_{12}O_6$



Ribose



Deoxyribose

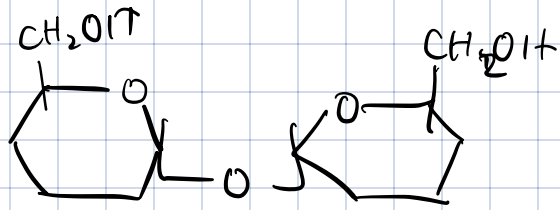


simplest sugar

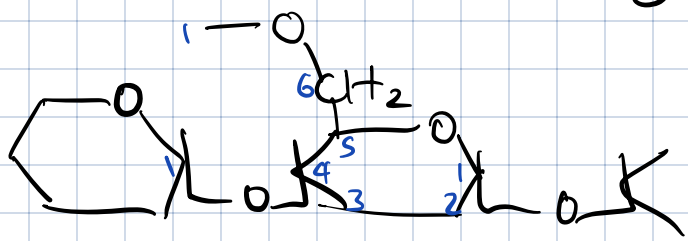
Polysaccharides

Amylose → straight chain

α 1–4 glycosidic bond



Amylopectin → branching



α 1–4 bond → straight

α 1–6 bond → branching

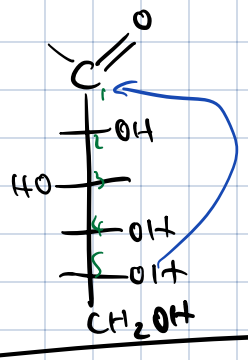
Glycogen → animal starch, highly branched

Cellulose → structural support

β 1–4 glycosidic bonds

Chitin

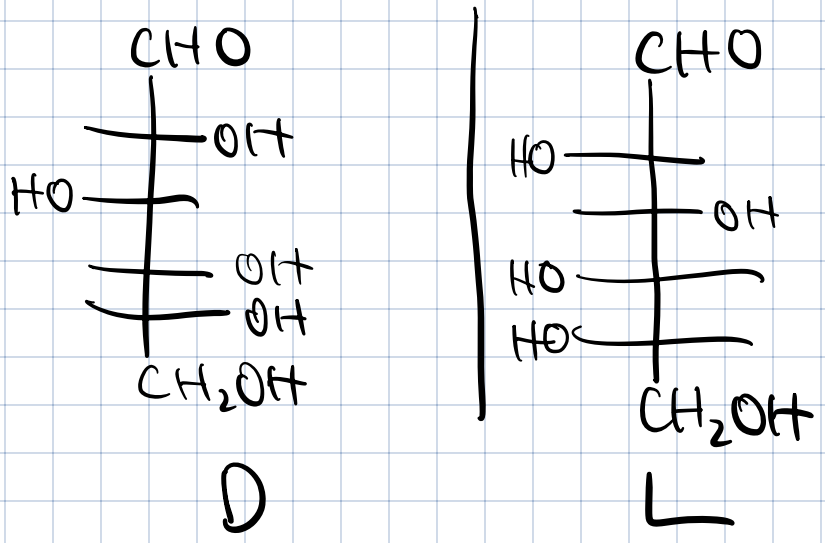
Cyclization



under mildly acidic conditions to activate aldehyde

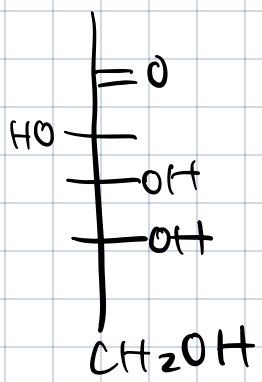
Stereochemistry of Glucose

D vs. L glucose: enantiomers

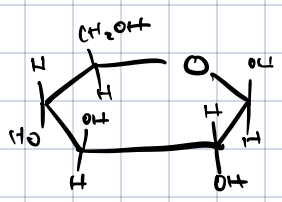


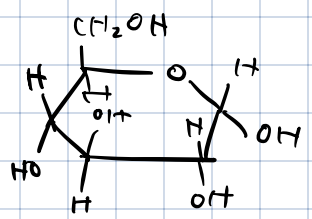
D L

Fructose



D-Fructose

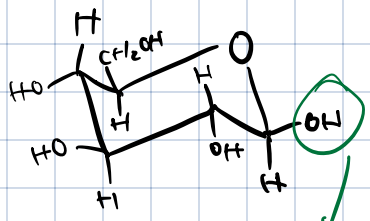




β-glucose α-glucose

In H_2O: 64% β, 36% α, <0.1% straight

Chair



β-anomer

This is why β is more favored → OH in eq

Glucose is pretty stable